

SIT742 2026T2 Assignment 2 Public Rubric

Forecasting Performance Determines Your Final Grade Band

Your Forecasting Performance Grade determines the grade band for your final A2 mark.

- **Forecasting measures:** The grade is determined primarily by MASE. MAPE is used as supporting cross-check evidence.
- **Questions Q1-Q7:** Your marks for these questions determine the exact final mark within the grade band.
- **Grade bands:** HD: 80-100; D: 70-79; C: 60-69; P: 50-59; F: 0-49.

For example, if your Forecasting Performance Grade is D, your final A2 mark will be between 70 and 79, with the exact mark determined by your marks for Questions Q1-Q7.

Purpose

This student-facing rubric summarises the public assessment expectations for Assignment 2. It should be read together with the [Assignment Specification](#), the [A2 Function Examples Guide](#), the [A2 Forecast Measures Guide](#), and the [Starter Notebook](#) that students complete.

Mark summary

Part	Questions	Focus	Marks
Part I	Q1-Q3	Forecast table generation and evaluation functions	15
Part II	Q4-Q6	Forecasting project, evaluation, and interpretation	70
Part III	Q7	Group video and collaboration presentation	15
		Total	100

How to read this rubric

This rubric summarises the public assessment expectations for each question. It should be read together with the Assignment Specification and completed using the Starter Notebook.

The mark allocations below show how each question is broadly assessed. They summarise public expectations rather than listing every possible test case, assessment scenario, or partial-credit case.

Forecast performance is assessed using both required measures: MASE with naive lag-1 scaling and MAPE with zero-actual handling, as explained in the [A2 Forecast Measures Guide](#). The two measures are not interchangeable: MASE assesses scaled error relative to historical movement, while MAPE assesses percentage error on valid nonzero actual rows. Performance thresholds and internal marker calibration details are not published.

Part I: Forecast Table Generation and Evaluation Functions [15 marks]

Q1: General naive-lag forecast table generator [5 marks]

Criterion	Marks	High achievement	Partial achievement	Needs improvement
Function signature and wide-format input/output	1	Defines <code>generate_naive_forecast_wide(...)</code> , accepts wide-format historical actual tables, and returns a pandas DataFrame with rows and columns in the required order.	Function is present but the signature, input assumptions, type, or required order needs review.	Required function is missing, renamed, or not assessable.
Configurable lag logic	1	Correctly supports positive integer lag, including <code>lag=1</code> recursive carry-forward logic and <code>lag=12</code> annual naive logic where available.	One lag mode is mostly correct but another is incomplete or unreliable.	Configurable lag behaviour is missing or mostly incorrect.
Fixed-cutoff implementation	1	Uses only actual values available at or before the supplied cutoff and does not use future actual/result values as forecast inputs.	Cutoff handling is plausible but needs clearer evidence or has a limited edge-case issue.	Forecast generation appears to use future actual/result values or unclear cutoff logic.
All-destination and arbitrary-horizon support	1	Supports all required destinations and arbitrary requested forecast months.	Works for some destinations or horizons but coverage is incomplete.	Does not support all-destination or multi-month forecasting.
Clean wide-format output	1	Output contains Date plus required destination columns only, with numeric forecasts and no <code>model_label</code> or diagnostic columns.	Output is mostly usable but one schema or value detail needs correction.	Output is not in the required wide forecast-table format.

Q2: Forecast and actual table validation/alignment [5 marks]

Criterion	Marks	High achievement	Partial achievement	Needs improvement
Function signature and audit structure	1	Defines <code>validate_forecast_actual_wide(...)</code> and returns the fixed 20-key Python audit dictionary specified for Q2.	Validator is present but the signature, dictionary type, required keys, or key order needs review.	Required validator is missing, renamed, or not assessable.
Forecast-only validation mode	1	With <code>actual_wide_df=None</code> , checks forecast schema readiness including required months, destination columns, duplicate dates, extra rows/columns, missing values, numeric conversion, and finite values.	Forecast-only checks are present but one or more important checks are incomplete.	Forecast-only validation is missing or unreliable.
Forecast-vs-actual alignment mode	1	When actual/result data is supplied, checks whether forecast and actual/result tables can align by Date and destination columns after using the required evaluation months.	Alignment checks are present but incomplete or difficult to inspect.	Forecast-vs-actual alignment checks are missing or unreliable.
Required month, destination, and value-quality checks	1	Checks required months, destination columns, numeric conversion, finite values, missing values, duplicate dates, and expected row counts.	Coverage or value-quality checks are present but incomplete.	Core coverage, duplicate-date, or value-quality checks are missing.
Useful audit output	1	Returns every required audit field with the specified meaning and mode-dependent None values.	Audit output is mostly complete but a required field, type, or mode-dependent value needs correction.	Audit output does not follow the required fixed contract.

Q3: Forecast accuracy measures from forecast and actual tables [5 marks]

Criterion	Marks	High achievement	Partial achievement	Needs improvement
Function signature and destination-level output	1	Defines <code>evaluate_forecast_wide(...)</code> and returns exactly (<code>destination_metrics</code> , <code>aggregate_metrics</code>) using the required DataFrame columns and dictionary keys.	Forecast-measure function is present but the signature, container types, required fields, or order needs review.	Required forecast-measure function is missing, renamed, or not assessable.

Criterion	Marks	High achievement	Partial achievement	Needs improvement
Required MASE and MAPE calculations	1	Correctly computes MASE and MAPE as separate destination-level measures on aligned evaluated rows, with MAE available to support MASE numerator checking.	Measures are mostly correct but one calculation or alignment detail needs review.	Required MASE or MAPE calculation is missing or mostly incorrect.
MASE denominator and diagnostics	1	Computes each destination denominator from all valid chronologically sorted lag-1 training pairs for that destination and reports denominator plus <code>n_denominator_pairs</code> .	Denominator logic is mostly correct but lagged-pair counting, sorting, or one edge case needs review.	MASE denominator is incorrect, mixes destination histories, or omits denominator diagnostics.
Aggregate measures	1	Returns the required unweighted aggregate keys <code>n_destinations</code> , <code>mean_mase</code> , <code>median_mase</code> , and <code>mean_mape</code> .	Aggregate measures are present but a required key or value needs review.	Required aggregate measures are missing or not inspectable.
Edge-case handling	1	Handles missing actuals, missing forecasts, zero actuals for MAPE, nonfinite values, and zero or unavailable denominators explicitly with <code>mase_available</code> and <code>denominator_warning</code> .	Edge-case handling is present but incomplete.	Important measure edge cases are not handled.

Part I integrated evidence

Part I integrated evidence does not carry separate marks. It supports Q1-Q3 and provides the minimum baseline evidence required in Q4.

It is not an additional mark item; missing or weak integrated evidence affects the relevant Q1-Q3 evidence and the Q4 baseline-comparison evidence.

Stronger submissions use Q1-Q3 together to create and visibly display `baseline_validation_comparison` and `baseline_validation_summary` for both `naive_lag1` and `naive_lag12`, cover all 20 destinations, evaluate validation months 2023M03 to 2023M07, and report both required measures, MASE and MAPE, alongside MAE, denominator, `n_denominator_pairs`, `mase_available`, and `denominator_warning`.

Part II: Forecasting Project, Evaluation, and Interpretation [70 marks]

Q4: Forecasting workflow and model development [20 marks]

Criterion	Marks	High achievement	Partial achievement	Needs improvement
Required baseline comparison	4	Includes required <code>naive_lag1</code> and <code>naive_lag12</code> validation baselines for all 20 destination series with validation MASE and MAPE evidence in the required minimum evidence-table schemas.	Baseline evidence is present but one required baseline, all-20 support, schema, or validation-measure detail is incomplete.	Required baseline comparison evidence is missing or not meaningful.
Improved model evidence	4	Includes at least one improved forecasting method with reproducible and stable evidence.	Improved model evidence is present but limited, hard to inspect, unstable on rerun, or not clearly compared.	Improved model evidence is missing.
Validation design and validation measures	4	Uses validation months 2023M03 to 2023M07, reports MASE with naive lag-1 scaling and MAPE with appropriate zero-actual handling, and avoids training/validation leakage.	Validation evidence is present but MASE/MAPE handling, cutoff handling, or coverage needs review.	Validation design or required MASE/MAPE evidence is missing or unreliable.
Model comparison and selection rationale	4	Justifies model choice by comparing the improved method against required baselines using validation evidence, destination behaviour, and limitations.	Model choice is plausible but baseline comparison, evidence linkage, or limitation discussion is incomplete.	Model selection is weakly justified or not compared against the required baselines.
Reproducibility, stability, and all-20 destination support	4	Workflow is organised, reproducible, stable on rerun where relevant, and supports all-20 destination forecasting and final wide-table creation.	Workflow is partly reproducible but some all-20, stability, or rerun evidence is weak.	Workflow is not reproducible, unstable, or does not support all-20 forecasting.

Q5: Final forecast submission and assessment-period performance evaluation [30 marks]

Criterion	Marks	High achievement	Partial achievement	Needs improvement
Required wide forecast object and CSV schema	4	forecast_submission_wide is visible in the notebook/PDF, forecast_submission_audit follows the fixed Q2 contract and passes forecast-only checks, and the submitted CSV uses Date plus exactly the 20 public-dataset destination columns with no extra columns.	Wide forecast schema or audit is mostly usable but one field, column, order, visibility, or export detail needs review.	Required wide forecast object, audit, or CSV schema is missing.
Required month coverage	4	Covers exactly 12 rows for 2023M08 to 2024M07 without duplicate dates or extra rows.	Month coverage is mostly complete but one month, duplicate, or extra-row issue needs review.	Required forecast month coverage is incomplete.
Numeric, finite, non-placeholder forecasts	4	Forecast values are numeric, finite, destination-specific, stable enough for assessment, and do not include assessment-period actual/result values.	Some forecast values need review for validity, stability, placeholder risk, or destination specificity.	Forecast values are missing, nonnumeric, nonfinite, placeholder-like, unstable, or include invalid actual/result information.
Assessment-period MASE and MAPE performance evidence	14	Submitted forecasts show strong performance on both required measures when compared with assessment-period actual/result values by the teaching-team evaluator, subject to valid schema and marker confirmation.	Performance evidence is assessable but one required measure is weaker, mixed across destinations, or requires manual review before confirming the score.	Required MASE or MAPE evidence cannot be validly computed or indicates weak forecast performance.
Forecast stability and destination-level sanity	4	Forecasts are plausible enough for marker review, destination-specific, stable under the documented workflow, and not obviously copied, constant placeholders, or structurally broken.	Forecasts are mostly plausible but need destination-level review, stronger stability evidence, or stronger justification.	Forecasts appear structurally broken, copied, constant, unstable, or implausible.

Exact assessment-period performance thresholds and internal marker calibration details are not published.

Q6: Performance-measure discussion and selected-market forecast analysis [20 marks]

Criterion	Marks	High achievement	Partial achievement	Needs improvement
Measure concepts: MASE and MAPE	4	Discusses MASE and MAPE, including strengths, weaknesses, and why they provide different required forecast-performance views.	Measure discussion is present but one measure or tradeoff needs more detail.	Measure discussion is missing or superficial.
Evidence use from validation measures and baseline comparison	6	Uses baseline_validation_summary, validation_metrics, and selected model evidence to support performance-measure discussion and model interpretation.	Uses some validation evidence but baseline comparison or interpretation links are incomplete.	Validation or baseline evidence is missing from the discussion.
Selected-market coverage	3	Analyses Australia, Japan, and two justified additional destinations with destination-specific evidence.	Selected-market discussion is present but one required or selected market is incomplete.	Selected-market analysis is missing or not destination-specific.
Per-market forecast analysis	5	For each selected market, discusses historical pattern, baseline comparison, selected model validation behaviour, final forecast shape and plausibility, and uncertainty or risks.	Per-market discussion is present but one evidence type or market-specific interpretation is incomplete.	Per-market forecast analysis is generic, unsupported, or missing.
Evidence presentation and report quality	2	Uses clear tables/figures, coherent narrative, and technically defensible conclusions.	Report is understandable but evidence presentation or narrative needs improvement.	Report quality or evidence presentation is weak.

Part III: Group Video and Collaboration Presentation [15 marks]

Q7: Group video and collaboration explanation [15 marks]

Criterion	Marks	High achievement	Partial achievement	Needs improvement
Video submission and duration	3	Video is submitted and normally within the 8-12 minute requirement, unless an approved arrangement exists.	Video is submitted but duration, access, or format needs review.	Video is missing or inaccessible.
All-member participation	3	Every listed group member participates or an approved arrangement is documented.	Participation is partly evident but needs review.	Member participation is missing or unclear.
Code execution and reproducibility demo	3	Video shows code execution and explains how submitted outputs, including the final forecast CSV, can be reproduced from the notebook.	Reproducibility demo is present but incomplete or does not clearly cover the final CSV.	Code execution or reproducibility demo is missing.
Technical explanation	3	Video explains problem, data, model, validation, forecast output, interpretation, and key results.	Technical explanation is present but incomplete or uneven across the workflow.	Technical explanation is missing or not connected to submitted work.
Contributions and collaboration	3	Video explains credible individual contributions and collaboration aligned with the submitted work.	Contribution explanation is present but underdeveloped or needs confirmation.	Contribution and collaboration explanation is missing or not credible.

Communication, reproducibility, and stability across the assignment

Communication, reproducibility, and stability are assessed within the relevant questions, especially Q4-Q7. They are not separate additional mark categories.

Across the assignment, stronger submissions generally:

- run from top to bottom in the submitted notebook;
- keep required object names and outputs easy to locate;
- set random seeds or document controlled stochastic variation where relevant;
- use labelled figures and tables;
- connect written explanations to generated evidence;
- acknowledge the TULIP Lab ISF-TDF2023 dataset and any external data sources;
- document cutoff-safe use of external data;
- export the final forecast CSV directly from `forecast_submission_wide` with `index=False`;
- avoid manual post-export editing of forecasts or reported measures;
- avoid local absolute paths, private files, credentials, restricted data, or undocumented dependencies.